DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: F. Y. B. Tech **Branch:** Common To All Branches **Semester:** I

Subject Code & Name: 24AF2PHYBS102, Engineering Physics

Max Marks: 60 **Date:** 08/02/2025 **Duration:** 3 Hr.

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	The speed of propagation of ultrasonic waves increases with increase in				Remember (CO1)	1
	a. Wavelength	b. Frequency	c. Amplitude	d. Intensity		
2	Dielectric materials are generally				Remember (CO1)	1
	a. Insulating Materials	b. Ferri Electric Materials	c. Ferro Electric Materials	d. Superconducting Materials		
3	In Newton's ring shape of interference pattern is				Remember (CO2)	1
	a. Straight fringes	b Circular fringes	c. Elliptical fringes.	d. Straight & Equidistant lines		
4	The substance which rotates the plane of polarization to left is called as				Remember (CO2)	1
	Dextrorotatory	b. Levorotatory	c. Oscillatory	d. None of these		
5	The principle of Laser is				Remember (CO2)	1
	a. Spontaneous emission	b. Stimulated emission	c. Thermionic emission	d. All of these		
6	Numerical aperture is also called as ____________ of the fiber				Remember (CO2)	1
	a. Reflecting angle	b. Sine of Acceptance angle	c. Scattering angle	d. Recoiling angle		
7	According to Heisenberg's principle, certainty in position involves				Remember (CO3)	1
	a. Uncertainty in momentum	b. certainty in momentum	c. uncertainty in position	d. certainty in position		
8	What is the fundamental unit of information in quantum computing				Remember	1

	a. Bit	b. Qubit	c. Byte	d. Quantum Byte	(CO3)	
9	Geiger Muller Counter is used to measure				Remember (CO4)	1
	a. α particles	b. β and ϒ particles	c. α, β & ϒ particles	d. None of these		
10	Number of atoms per unit cell for Face centered Cubic structure is				Remember (CO4)	1
	a.1	b. 4	c. 2	d. 6		
11	The temperature at which normal material turns into superconductor is				Remember (CO5)	1
	a. Absolute Temperature	b. Critical Temperature	c. Mean Temperature	d. Crystallization Temperature		
12	1 Nanometer=________m				Remember (CO5)	1
	a. 10^{9}	b. 10^{-10}	c. 10^{-9}	d. 10^{10}		
Q. 2	**Solve the following.**					**12**
A)	What is Piezoelectric effect? Describe the production of ultrasonic waves by using Piezoelectric method.				**Remember/ Understand (CO1)**	**6**
B)	Explain any three factors affecting architectural acoustics of a building. A cinema hall has a volume of 7500 m^3 . It is required to have reverberation time of 1.5 sec. What should be the total absorption in the hall?				**Understand (CO1)**	**6**
Q.3	**Solve the following.**					**12**
A)	Derive an expression for diameter of n^{th} bright and dark Newton's rings.				**Understand (CO2)**	**6**
B)	Explain the construction and working of Helium Neon laser.				**Understand (CO2)**	**6**
Q. 4	**Solve Any Two of the following.**					**12**
A)	What is Heisenberg's uncertainty principle? If the uncertainty in position of an electron is $4X10^{-10}$m. Calculate the uncertainty in its momentum.				**Remember/ Understand (CO3)**	**6**
B)	Derive time independent Schrodinger wave equation.				**Understand (CO3)**	**6**

C)	Derive time dependent Schrodinger wave equation.	**Understand (CO3)**	**6**
Q.5	**Solve Any Two of the following.**		**12**
A)	Define atomic packing fraction. Calculate the atomic packing fraction in SC, BCC, FCC lattices.	**Remember/ Understand (CO4)**	**6**
B)	Derive the relation between lattice parameter 'a' and crystal density 'ρ' Copper has FCC structure and its atomic radius is 1.278 $\mathring{A}$. Calculate density of Cu. Given atomic weight of Cu=63.5.	**Understand (CO4)**	**6**
C)	With neat diagram explain the construction & working of Geiger Muller Counter.	**Understand (CO4)**	**6**
Q. 6	**Solve Any Two of the following.**		**12**
A)	Explain the B-H curve for ferromagnetic materials. Define Coercivity and retentivity	**Understand (CO5)**	**6**
B)	Define superconductivity and distinguish between Type I & Type II superconductors.	**Understand (CO5)**	**6**
C)	What is nanomaterial? Explain top-down and bottom-up approach for synthesis of nanomaterial	**Understand (CO5)**	**6**
	***** End *****		